



Construction Site Management Platform

DBMS Project-Based Learning – Review 3

PRESENTED BY

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From Database to Working Application

Key Capabilities

- Project and site management
- Procurement and delivery tracking
- Material stock management
- Work package and progress tracking
- Contractor billing and payment management
- Reports and database-driven insights
- CRUD operations with validation

Application Architecture

User

BUILDREC UI

React + TypeScript

Node.js + Express API

MySQL Database

Database Design

Database Implementation

BUILDREC Application

Working UI + CRUD + Reports

Technology & Implementation

DATABASE

MySQL 8.0+

BACKEND

Node.js + Express +
TypeScript

FRONTEND

React + TypeScript
+ Vite

CONNECTIVITY

mysql2 / REST API

Implementation Highlights

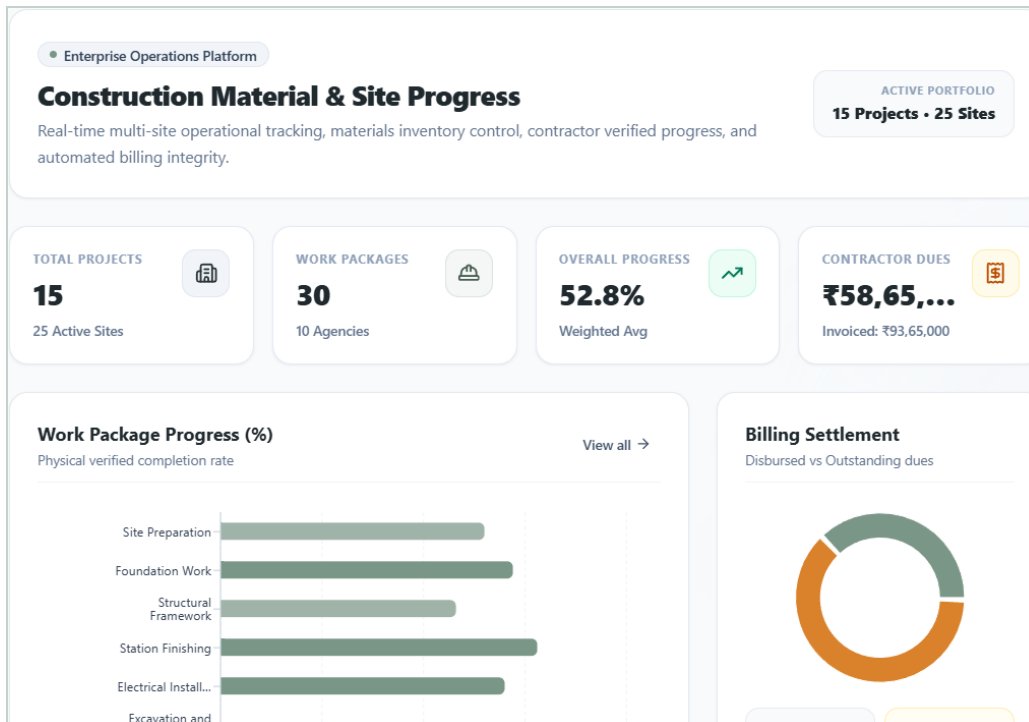
- 15 normalized database tables
- Primary and foreign key constraints
- NOT NULL, CHECK and referential integrity

Business Logic & UI

- Database triggers for business-rule enforcement
- Analytical SQL views for reporting
- Responsive UI for desktop, tablet and mobile

Dashboard Shows

- Active Projects
- Registered Sites
- Live Material Stock
- Overall Progress
- Total Billed vs. Paid
- Pending Contractor Payments
- Material Consumption
- Project-wise Financial Overview
- Recent Activity



Projects, Sites & Procurement

Projects & Sites

- View and search project records
- Manage construction sites under projects
- Add new projects and sites
- Filter sites by project
- Maintain project-site relationships

Procurement & Deliveries

- Manage suppliers
- Create purchase orders
- Record site deliveries
- Add delivery items and quantities
- Track incoming materials

PROJECT ID	PROJECT NAME	ASSOCIATED SITES	ACTIONS
P181	Metro Rail Station	2 Sites	+ Site Delete
P182	Green Valley Apartments	2 Sites	+ Site Delete
P183	City Hospital Complex	1 Site	+ Site Delete
P184	Tech Park Phase 2	1 Site	+ Site Delete
P185	Riverfront Commercial Center	2 Sites	+ Site Delete

DELIVERY ID	PO REF	DESTINATION SITE	DELIVERY DATE	ACTIONS
1896	PO #701	North Station Site	2025-09-04	+ Receive Items Delete
1897	PO #702	North Station Site	2025-08-06	+ Receive Items Delete
1898	PO #703	Tower A Site	2025-08-08	+ Receive Items Delete
1894	PO #704	Tower B Site	2025-09-10	+ Receive Items Delete
1895	PO #705	Main Hospital Site	2025-09-12	+ Receive Items Delete
1899	PO #706	Tech Park Main Site	2025-08-14	+ Receive Items Delete

Materials, Stock & Progress

Materials & Site Stock

- Manage material catalogue
- View stock by site and material
- Track available quantities
- Issue materials to work packages
- Enforce stock availability

STOCK ID	SITE LOCATION	MATERIAL NAME	AVAILABLE QUANTITY	ACTIONS
#1001	North Station Site	Cement	1200 Bags (50kg)	Issue Edit Delete
#1002	North Station Site	Sand	1000 Cubic Meters	Issue Edit Delete
#1003	North Station Site	Steel	1000 Metric Tons	Issue Edit Delete
#1004	South Station Site	Electrical Cable	500 Meters	Issue Edit Delete
#1005	South Station Site	Cement	1100 Bags (50kg)	Issue Edit Delete
#1006	Tower A Site	Cement	1500 Bags (50kg)	Issue Edit Delete

Work Packages & Progress

- View work packages and contractors
- Log progress entries
- Assign labour teams
- Track completion percentage
- Validate progress between 0–100%

PACKAGE ID	PACKAGE NAME	SITE LOCATION	CONTRACTOR	VERIFIED QUANTITY	ACTIONS
#4001	Site Preparation	North Station Site	BuildRight Constructors	800 units	Log Progress Edit Delete
#4002	Foundation Work	North Station Site	BuildRight Constructors	1200 units	Log Progress Edit Delete
#4003	Structural Framework	North Station Site	Apex Infrastructure	1500 units	Log Progress Edit Delete
#4004	Station Finishing	South Station Site	Apex Infrastructure	900 units	Log Progress Edit Delete

Billing & Payments

Contractor Billing

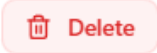
- View contractor bills
- Track billed quantities
- Verify work-package progress
- Control invoice creation
- Track outstanding balances

Payments

- Process payments against bills
- Track payment history
- Prevent payments exceeding bill
- Auto-reflect settlement status

Database Validation

- Billing requires verified progress
- Payment cannot exceed billed amount
- Triggers enforce financial rules

BILL ID	WORK PACKAGE	BILL AMOUNT	PAID TOTAL	BALANCE DUE	BILL DATE	ACTIONS
#1401	Site Preparation	₹7,50,000	₹5,00,000	₹2,50,000	2026-09-08	 Pay Now   Delete

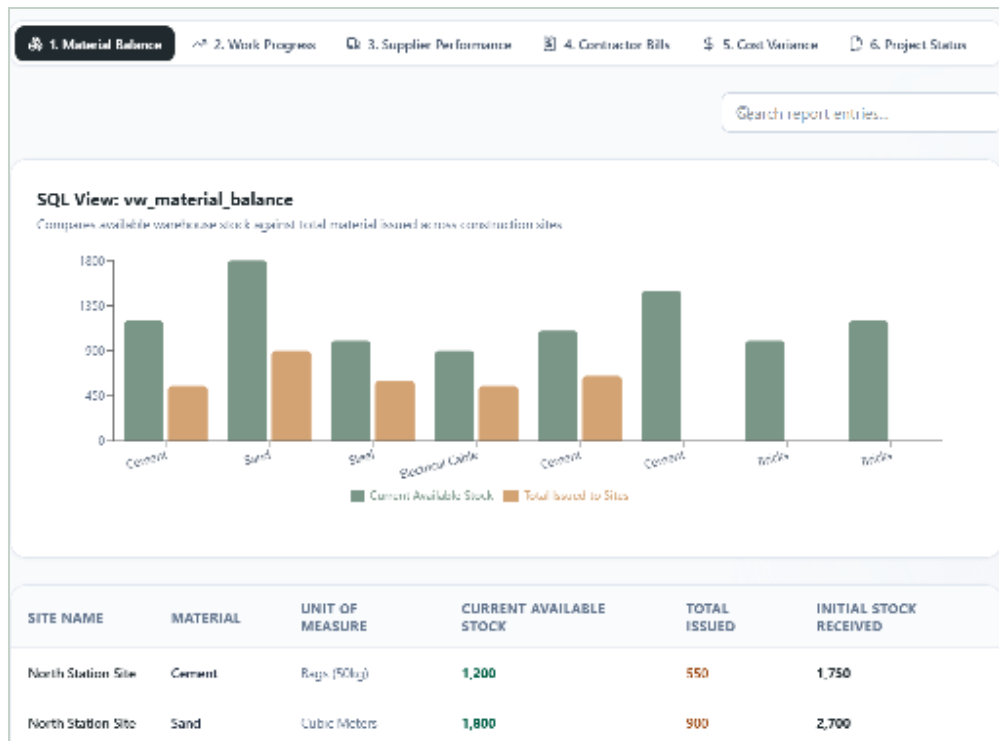
Database-Driven Reports

Reports

- Material Balance
- Work Progress
- Supplier Performance
- Contractor Bills & Balances
- Project Cost Variance · Project Status

Analytics

- Material availability and consumption
- Work-package completion
- Supplier delivery performance
- Contractor billing and payments
- Project financial summaries



CRUD & Database Validation

INSERT

Add a project / site / record

VIEW

Retrieve live database records

DELETE

Remove a record with
confirmation

VERIFY IN MYSQL

Confirm in MySQL Workbench

CRUD Operations

Insert
View
Delete
Search

Validation

Progress outside 0–100% →Rejected
Material issue above stock →Rejected
Billed qty above verified work →Rejected
Invalid delivery date →Rejected
Payment above bill amount →Rejected
Foreign keys protect related records

Conclusion & Future Scope

Conclusion

- Centralized construction project data
- Integrated material and site management
- Progress and contractor tracking
- Billing and payment management
- CRUD operations with validation
- Database-driven reports
- Real-time application–database integration

Future Scope

- Role-based authentication
- Multi-user access control
- Advanced project analytics
- Mobile/PWA deployment
- Cloud deployment
- Automated alerts and notifications
- Integration with BIM/site monitoring

BUILDREC converts the normalized database into a working site management application connected to MySQL.

Thank You